For complex vector spaces, the triangle inequality takes a slightly more complicated form. Let |*X*> and |*Y*> be any two vectors in a complex vector space. The magnitudes of the three vectors |*X*>, |*Y*>, and |*X*> + |*Y*> are

$$|X| = \sqrt{\langle X|X\rangle}$$

$$|Y| = \sqrt{\langle Y|Y\rangle}$$

$$|X+Y| = \sqrt{(\langle X| + \langle Y|)(|X\rangle + |Y\rangle)} \qquad (5.8)$$

We now follow the same steps as we did for the real case: First write

$$|X| + |Y| \geq |X+Y|.$$

Then square it and simplify:

$$2|X||Y| \geq |\langle X|Y\rangle + \langle Y|X\rangle|. \qquad (5.9)$$

This is the form of the Cauchy-Schwarz inequality that will lead to the uncertainty principle. But what does it have to do with the two observables **A** and **B**? We'll find out by cleverly defining |*X*> and |*Y*>.

5.7 THE GENERAL UNCERTAINTY PRINCIPLE

Let |Ψ> be any ket and let **A** and **B** be any two observables. We now define |*X*> and |*Y*> as follows:

$$|X\rangle = \mathbf{A}|\Psi\rangle$$

$$|Y\rangle = i\mathbf{B}|\Psi\rangle. \qquad (5.10)$$